

CodePulse

Group Coding Accountability Platform

Product Requirements Document

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1. Executive Summary

CodePulse is a web-based group accountability platform designed for computer science students. It connects to users' WakaTime accounts to pull real coding activity data, organises users into groups, and displays individual and group-level statistics on a shared leaderboard dashboard. The platform sends intelligent SMS reminders based on each user's historical coding patterns and self-declared goals, helping students build consistent coding habits through peer visibility and data-driven nudges.

2. Problem Statement

CS students often struggle with maintaining consistency in coding practice. While existing tools like WakaTime provide individual activity tracking, they offer no peer visibility or accountability layer. There is currently no free, dedicated tool that combines real coding data with group motivation, goal tracking, and smart reminders tailored to individual habits.

3. Goals and Objectives

- Enable groups of CS students to share their coding activity in a transparent, motivating environment.
- Allow users to set language-specific or time-based coding goals and track progress against them.
- Leverage each user's historical WakaTime data to deliver reminders at the most effective time — not too early, not after the coding window has already passed.

- Provide granular privacy controls so users decide exactly what data to share with their group.

4. Tech Stack

Layer	Technology
Frontend	Next.js
Backend & Database	Supabase (PostgreSQL, Auth, Edge Functions)
SMS Integration	Africa's Talking or Twilio
Authentication	Google OAuth via Supabase Auth

5. User Roles

Role	Description
Group Admin	Creates the group, receives the group ID, manages members
Member	Joins via group ID, connects WakaTime, sets goals, views dashboard

6. Core Features

6.1 Authentication

- Sign up and log in using Google (Gmail) OAuth only.
- No email/password registration is supported in v1.

6.2 Group Management

- Any authenticated user can create a group and automatically becomes its admin.
- A unique group ID is generated upon creation and shared with prospective members.
- Members join by entering the group ID after logging in.
- Admins can remove members from the group at any time.

6.3 WakaTime Integration & Onboarding

After joining a group, users are guided through a step-by-step onboarding flow:

1. Create a WakaTime account (if they do not already have one).
2. Install the WakaTime extension in their preferred IDE.
3. Connect their WakaTime API key to CodePulse.

Once connected, CodePulse syncs coding data on a scheduled basis (minimum every 30 minutes).

6.4 Group Dashboard

- Displays a leaderboard of all group members ranked by coding time, filterable by Today / This Week / This Month.
- Each member's card shows: display name, total time coded, top programming language, and top project.
- A bar chart visualises relative coding time per member for quick at-a-glance comparison.
- Privacy controls allow each user to hide specific data fields (e.g., hide project names while still showing total time).

6.5 Goal Setting

- Users can create personal goals, for example: "Code in Java for at least 1 hour today."
- Goals specify: programming language (optional), minimum duration, and deadline (daily or weekly).
- WakaTime data is automatically checked against each goal to determine completion status.
- Completed goals are marked with a visual indicator visible to all group members.

6.6 Intelligent SMS Reminders

- The system analyses each user's historical WakaTime data to identify their typical coding hours.
- If a goal deadline is approaching and the user has not yet met their target, an SMS reminder is sent.
- Reminder timing is optimised to arrive when the user is still within their usual coding window — neither too early nor after the window has already closed.
- Users can opt out of SMS reminders at any time from their profile settings.

7. User Stories

ID	As a...	I want to...	So that...
US-01	Student	Sign in with my Gmail	I don't need another password

US-02	Student	Create a group and share the ID	My friends can join
US-03	Member	Connect my WakaTime account	My coding data is visible to the group
US-04	Member	See a leaderboard of my group	I know how I compare
US-05	Member	Set a goal to code in Python for 30 mins	I stay accountable
US-06	Member	Receive an SMS when I haven't met my goal	I get reminded before it's too late
US-07	Member	Hide my project names from the group	I can protect sensitive work
US-08	Admin	Remove an inactive member	The group stays relevant

8. Non-Functional Requirements

- **Performance:**
- **Performance:** Dashboard must load within 3 seconds for groups of up to 50 members.
- **Data Freshness:** WakaTime data must be synced at least every 30 minutes.
- **Reliability:** Concurrent user requests handled reliably via Supabase's managed infrastructure.
- **Security:** All data transmissions must use HTTPS. Users' WakaTime API keys must be stored encrypted at rest.

9. Out of Scope (v1.0)

- Mobile application (iOS / Android)
- Multiple group memberships per user (v1 supports one group per user)
- Public group profiles or discovery
- In-app messaging or chat
- GitHub / GitLab repository integration

10. Success Metrics

Metric	Target
WakaTime Connection Rate	≥80% of members connect within 48 hours of

	joining
Active Coding Frequency	Average ≥ 3 active coding days per week per member
SMS Engagement	Open/action rate above 40%
User Motivation	Positive feedback via in-app prompt after 2 weeks

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